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Project Title:	Circadian Entrainment, Sleep-Wake Regulation and Neurobehavioral Performance During Extended Duration Space Flight		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Human Performance Factors, Sleep, and Chronobiology Team		
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Space Biology Element:	None		
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Zip Code:	02115-5804	**Congressional District:**	8
Comments:			
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
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Task Description:	NASA's planned exploration class mission to Mars will require crewmembers to adapt to a 24.65-h solar day throughout their >540 day stay on Mars. The ability of astronauts to sustain a high level of performance during their mission will be critically dependent upon adaptation of their circadian pacemaker to the 24.65-h day. Our data suggest that most astronauts would exhibit circadian misalignment if the space flight lighting conditions of <25 lux on the windowless middeck of the space shuttle were present during their stay on Mars. Circadian misalignment would result in sleep impairment, endocrine disturbances, and impaired neurobehavioral function. Preliminary data from our laboratory reveal that intermittent bright light exposure is effective in maintaining entrainment of the circadian pacemaker to longer-than-24 h days. They also suggest that lighting levels of ~25 lux - ~100 lux are insufficient to maintain a normal phase angle of entrainment. Other preliminary results demonstrate that blue light (460 nm) is more efficient than white light in resetting the human circadian pacemaker. Given that bright light pulses might not of practical use on Mars—due to time and energy constraints—these results demonstrate the need to develop practical and cost-effective countermeasures to adapt the human circadian pacemaker of all crewmembers to the 24.65-h day. In the current grant we proposed to test a countermeasure of "blue-enriched" light to entrain the human circadian pacemaker to a 24.65-h day. Specifically, we proposed to test the following hypotheses: i) that synchronization of the human circadian pacemaker to a 24.65-h day initiated at an adverse phase will not be appropriate in the presence of ~100 lux of white light; ii) that inappropriate circadian synchronization will result in the secretion of the sleep-promoting hormone melatonin during waketime, abnormal somatotropic and corticotropic activity, disturbed sleep, and impaired performance and daytime alertness; and iii) that exposure to "blue enriched" light during the daytime will establish a normal entrained circadian phase in subjects scheduled to a 24.65-h day. Seven healthy subjects participated in a 71-day inpatient protocol consisting of 3 baseline days (16:8 h wake:sleep) followed by 60 experimental days (24.65 h, 16.43:8.22 h). Light intensity during the experimental days was ~ 100 lux (n=2) and ~50 lux (n = 5) at 137 cm height in the horizontal angle. Circadian phase was measured by the Dim Light Melatonin Onset (DLMO) measured 6 times under dim light (~ 1.8 lux at 137 cm height in the horizontal angle) and postural control (before and after the experimental days, and on experimental Days 6, 20, 34, & 48) and during several segments on the experimental days. The critical manipulation in the study was the timing of sleep on the first experimental day was shifted 12 h compared to baseline; equivalent to a slam shift. To assess entrainment, following the “slam shift” phase angle was calculated from the difference between DLMO and scheduled bedtime. The results showed that the average phase angle for subjects on the first 24.65h day was 12.5 + .26 hours for the 50 lux group and 8.96 + .96 hours for the 100 lux group, which is indicative of acute circadian misalignment. The average phase angle on the 6th experimental day was 5.8 + 1.6h for the 50 lux group and 7 + 3.7 h for the 100 lux group, showing that subjects continued to be misaligned. The average phase angle on the 15th experimental day was 1.5 + .46. Thereafter they stabilized. The average phase angle was 2.4 + .31h on experimental day 34 and 2.2 + .26h on experimental day 48 (p>.05). The analysis of the cognitive performance data under these conditions is ongoing. It is important to note the following about these results. First, our results show that crew members would be most vulnerable to the detrimental effects of circadian misalignment during the first two weeks of a mission. This is important, because the duration of short-term mission such as those to the ISS and the moon are approximately two weeks long. Second, the data reveal that light intensities that are similar to the lighting conditions during space flight are insufficient to induce rapid entrainment during a short term mission. Therefore, we feel that it is critical to develop countermeasures that would facilitate sleep and circadian adaptation rapidly. Based on our current results and based on the Vision for Space exploration outlined by President Bush, we feel that it is important to refocus our research efforts on the circadian and sleep-wake challenges associated with short duration lunar exploration. To do so, we plan to modify our studies during the remainder of this grant period to explore this initial period of vulnerability further and specifically test the effectiveness of our proposed countermeasure (blue enriched light) to hasten circadian adaptation to short term missions and thereby improve sleep during scheduled sleep episodes and improve performance during scheduled wake episodes. The results of the proposed studies will answer fundamental questions on the mechanisms underlying circadian entrainment in humans and could have a profound effect on the health, productivity and safety of astronauts during space exploration missions.The current project, aimed at evaluating the effectiveness of blue-enriched light exposure over white light as a countermeasure for circadian misalignment during planned missions to the ISS, Moon and further destinations (as outlined in the Vision for Space exploration)using the proposed Crew Exploration Vehicle (CEV) will help us address these key issues.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The purpose of the proposed studies is to test the updated specific hypotheses aimed at evaluating the effectiveness of blue-enriched light exposure over white light as a countermeasure for circadian misalignment during exploraation class space missons. Results from previous studies in our lab indicate exposure to blue-enriched light will be a more effective means of entraining the human circadian pacemaker compared to exposure to white light of similar photon density, or even to light of greater photon density but of inappropriate spectral content (long wavelengths). The funded research project has important implications for earth-based applications, particularly in the development of treatment for circadian rhythm disorders. This project, holds the potential for producing shorter, more efficient light treatment regimens, which would not only increase compliance in clinical populations (e.g. night shift workers, people with Advanced and Delayed Sleep Phase Syndromes), but would make light treatment more practical in industrial/work settings.
	In the current grant we proposed to test a countermeasure of "blue-enriched" light to entrain the human circadian pacemaker to a 24.65-h day. Seven healthy subjects participated in a 71-day inpatient protocol consisting of 3 baseline days (16:8 h wake:sleep) followed by 60 experimental days (24.65 h, 16.43:8.22 h). Light intensity during the experimental days was ~ 100 lux (n=2) and ~50 lux (n = 5) at 137 cm height in the horizontal angle. Circadian phase was measured by the Dim Light Melatonin Onset (DLMO) measured 6 times under dim light (~ 1.8 lux at 137 cm height in the horizontal angle) and postural control (before and after the experimental days, and on experimental Days 6, 20, 34, & 48) and during several segments on the experimental days. The critical manipulation in the study was the timing of sleep on the first experimental day was shifted 12 h compared to baseline; equivalent to a slam shift. To assess entrainment, following the “slam shift” phase angle was calculated from the difference between DLMO and scheduled bedtime.

Task Progress:	We have so far analyzed the plasma melatonin data. To assess entrainment, phase angle was calculated from the difference between DLMO and scheduled bedtime. The results showed that the average phase angle for subjects on the first 24.65h day was 12.5 + .26 hours for the 50 lux group and 8.96 + .96 hours for the 100 lux group, which is indicative of acute circadian misalignment. The average phase angle on the 6th experimental day was 5.8 + 1.6h for the 50 lux group and 7 + 3.7 h for the 100 lux group, showing that subjects continued to be misaligned. The average phase angle on the 15th experimental day was 1.5 + .46. Thereafter they stabilized. The average phase angle was 2.4 + .31h on experimental day 34 and 2.2 + .26h on experimental day 48 (p>.05). The Analysis of the cognitive performance data under these conditions is ongoing. It is important to note the following about these results. First, our results show that crew members would be most vulnerable to the detrimental effects of circadian misalignment during the first two weeks of a mission. This is important, because the duration of short-term mission such as those to the ISS and the moon are approximately two weeks long. Second, the data reveal that light intensities that are similar to the lighting conditions during space flight are insufficient to induce rapid entrainment during a short term mission. Therefore, we feel that it is critical to develop countermeasures that would facilitate sleep and circadian adaptation rapidly.
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